

IEAP-python-pre-series02

Vidusha Sameen Thebuwana

1. What is a sinusoid ?

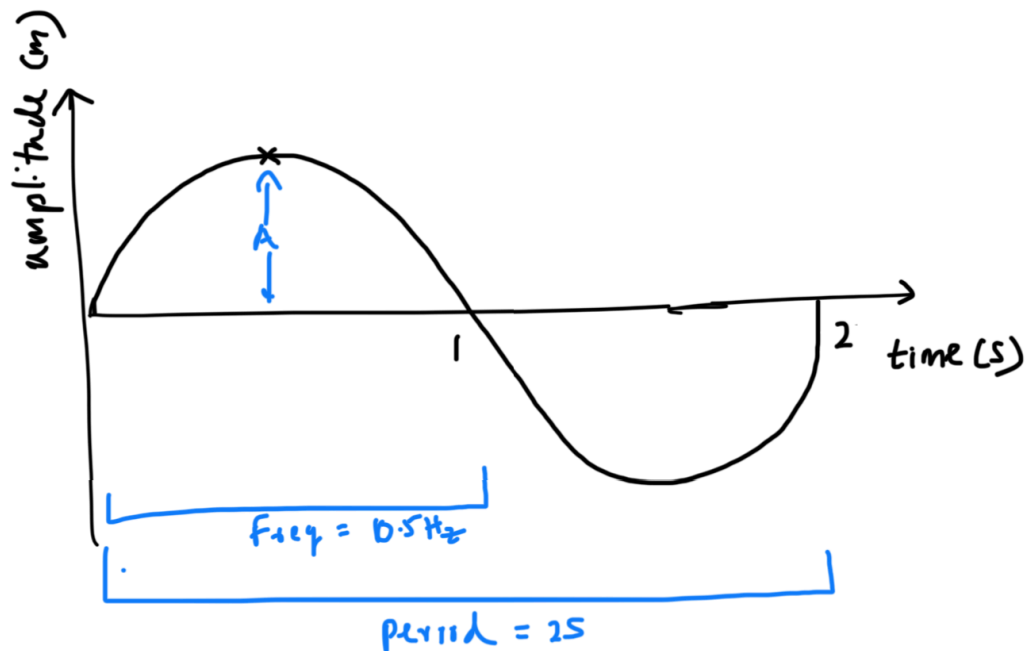
A sinusoid is a harmonic wave that moves in a periodic manner with smooth edges and oscillates between a baseline.

<https://www.sciencedirect.com/topics/chemistry/sinusoidal-wave>

2. What (simple) physical phenomenon produces a sinusoid?.

A tuning fork has the ability to create pure tones (sinus waves)

3. Draw a diagram of a sine curve as a function of time



4. What is the equation of a sinusoid?

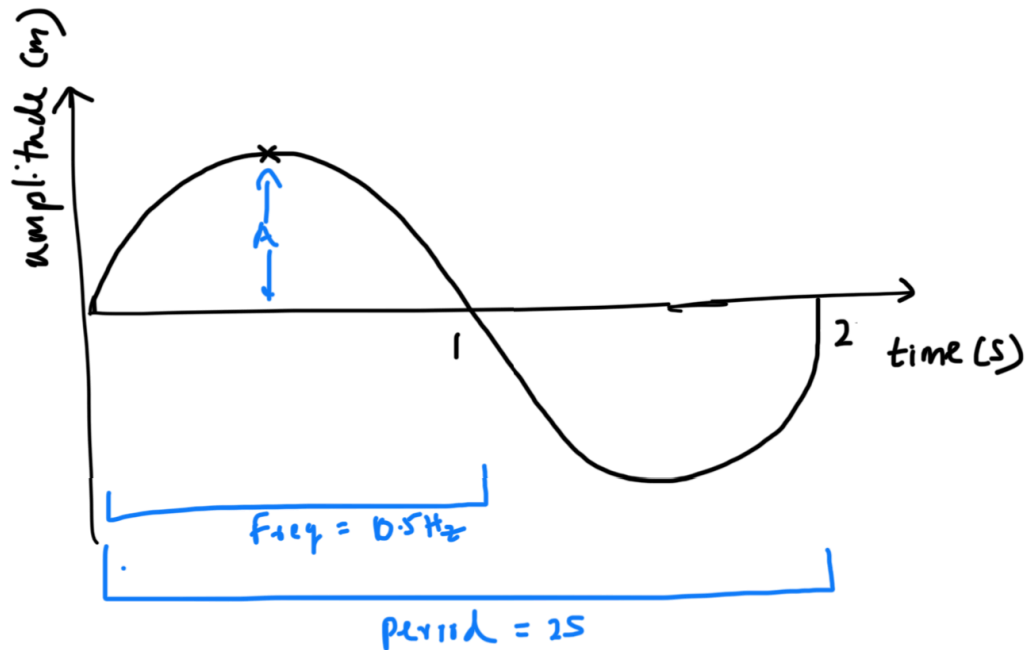
$$y(t) = A \sin(\omega t + \phi)$$

t = time (Seconds)

A = Amplitude (the peak Deviatiton from zero)

ϕ = Initial phase angle (radians, rad)

5. Define (in plain English + give unit) and illustrate (on the diagram):



frequency - oscillations per given time (Hz)

period - time taken for one oscillation (Seconds)

amplitude - Peak deviation from the baseline (Meters)

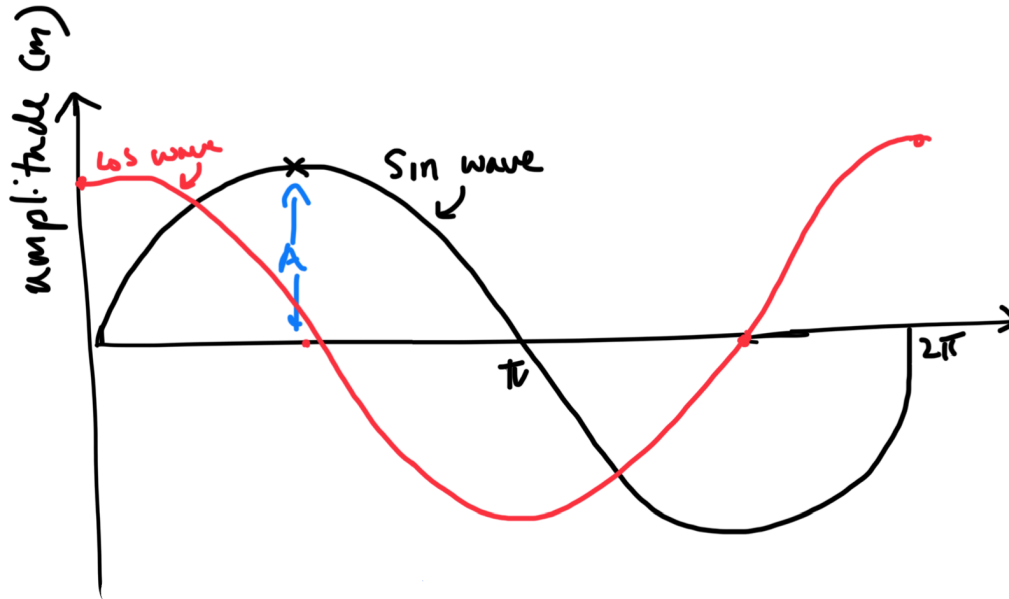
pulsation/ angular frequency - the speed of the wave (rad/s)

<https://www.electronics-tutorials.ws/accircuits/sine-wave.html>

6. What is the difference between a sine and a cosine?

The core difference between a sine wave and a cosine wave is their starting position at $x=0$, also known as their phase. Both functions share the exact same shape, period (2π radians), and amplitude range (from -1 to 1), but cosine is simply a sine wave shifted horizontally.

$$y \cos(\omega t) = A \cos(\omega t)$$



Gemini: "what is the difference between sin wave and cos wave. please provide the functions"

7. Explain why a sinusoid is a periodic signal

A sinusoid is periodic because it repeats its values continuously at regular time intervals T

[https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Signal_Processing_and_Modeling/Signals_and_Systems_\(Baraniuk_et_al.\)/06%3AContinuous_Time_Fourier_Series_\(CTFS\)/6.01%3AContinuous_Time_Periodic_Signals](https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Signal_Processing_and_Modeling/Signals_and_Systems_(Baraniuk_et_al.)/06%3AContinuous_Time_Fourier_Series_(CTFS)/6.01%3AContinuous_Time_Periodic_Signals)

8. Explain why a sinusoid is a continuous signal

Because it is defined smoothly across all real-valued time points without sudden breaks or step changes

9. Explain why a sinusoid is a parametric signal, and with which parameters

Because its full mathematical form is completely governed by three explicit parameters: amplitude, frequency, and phase shift

gemini: "Explain why a sinusoid is a parametric signal, and with which parameters"

10. Provide at least 3 URL and 3 references to books or scientific papers that you have used to answer the previous questions

1. <https://www.electronics-tutorials.ws/accircuits/sine-wave.html>
2. <https://www.sciencedirect.com/topics/chemistry/sinusoidal-wave>
3. https://en.wikipedia.org/wiki/Sine_and_cosine

4. https://en.wikipedia.org/wiki/Sine_wave
5. [https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Signal_Processing_and_Modeling/Signals_and_Systems_\(Baraniuk_et_al.\)/06%3A_Continuous_Time_Fourier_Series_\(CTFS\)/6.01%3A_Continuous_Time_Periodic_Signals](https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Signal_Processing_and_Modeling/Signals_and_Systems_(Baraniuk_et_al.)/06%3A_Continuous_Time_Fourier_Series_(CTFS)/6.01%3A_Continuous_Time_Periodic_Signals)
11. Provide the time you spent on this assignment - 2.5 hours